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GOVERNMENT OF INDIA

**OFFICE OF DIRECTOR GENERAL OF CIVIL AVIATION**

TECHNICAL CENTRE, OPP SAFDARJANG AIRPORT, NEW DELHI

**CIVIL AVIATION REQUIREMENTS**

**SECTION 2 - AIRWORTHINESS**

**SERIES I PART V**

**ISSUE III, 30<sup>th</sup> October 2018**

**EFFECTIVE: FORTHWITH**

F.No. DGCA-25012(07)/5/2021-AW

**SUBJECT: FLIGHT DATA RECORDERS, COMBINATION RECORDERS, DATALINK RECORDERS, AIRBORNE IMAGE RECORDERS, AIRBORNE IMAGE RECORDING SYSTEM AND AIRCRAFT DATA RECORDING SYSTEM**

**1. PURPOSE:**

Rule 57 of Aircraft Rules, 1937 requires that every aircraft shall be fitted and equipped with instruments and equipment, including radio apparatus and special equipment as may be specified according to the use and circumstances under which the flight is to be conducted.

This part of Civil Aviation Requirement lays down the requirements for fitment of Flight Data Recorders, Combination Recorders, Data link Recorders, Airborne Image Recorders, Airborne Image Recording System and Aircraft Data Recording System on aircraft registered in India. All aircraft imported/ purchased or leased for operation in India shall meet the applicability requirements laid down in this CAR.

This CAR has been issued under the provisions of Rule 29C of the Aircraft Rules, 1937.

**2. DEFINITIONS:**

**Flight Recorder:** Any type of recorder installed in the aircraft for the purpose of complementing accident/incident investigation.

**Automatic deployable flight recorder (ADFR):** A combination flight recorder installed on the aircraft which is capable of automatically deploying from the aircraft.

**Commercial Operation:** An aircraft operation involving the transport of passengers,

cargo or mail for remuneration or hire.

**General Aviation:** An aircraft operation other than a commercial air transport operation or an aerial work operation.

### 3. FLIGHT RECORDER COMPOSITION:

- 3.1 Crash protected flight recorders comprise one or more of the following Systems: a flight data recorder (FDR), a cockpit voice recorder (CVR), an airborne image recorder (AIR), a data link recorder (DLR). Image and data link information may be recorded on either the CVR or the FDR.

When image or data link information is required to be recorded on a crash-protected flight recorder, it is permissible to record it on either the CVR or the FDR.

- 3.2 Light weight flight recorders comprise one or more of the following:

An aircraft data recording system (ADRS), a cockpit audio recording system (CARS), an airborne image recording system (AIRS), a data link recording system (DLRS).

When image or data link information is required to be recorded on a lightweight flight recorder, it is permissible to record it on either the CARS or the ADRS.

**Note 1-** *For aeroplanes / helicopters for which the application for type certification is submitted before 1 January 2016, specifications applicable to crash protected flight recorders may be found in EUROCAE ED- 112, ED-56A, ED-55, Minimum Operational Performance Specifications (MOPS), or earlier equivalent documents.*

**Note 2-** *For aeroplanes / helicopters for which the application for type certification is submitted on or after 1 January 2016, specifications applicable to crash protected flight recorders may be found in EUROCAE ED-112A, Minimum Operational Performance Specification (MOPS), or equivalent documents.*

**Note 3 -** *Specifications applicable to lightweight flight recorders may be found in EUROCAE ED 155, Minimum Operational Performance Specification (MOPS), or equivalent documents.*

**Note 4-** *Detailed requirements on flight data recorders are contained in Appendix-I.*

## 4 Applicability

### 4.1 Aeroplane– Commercial Air Transport

- 4.1.1 ~~All turbine~~–engined aeroplanes, for which the individual certificate of airworthiness was first issued before 1 January 1987, with a maximum certificated take-off mass of over 27000kg that are of types of which the prototype was certificated by the appropriate national authority after 30 September, 1969 should be equipped with an FDR which should record, in addition to the first 5 parameters listed in table 1 of Appendix I, such additional parameters as are necessary to meet the objective of determining:
- a) the attitude of the aeroplane in achieving its flight path; and
  - b) the basic forces acting upon the aero plane resulting in the achieved flight path and the origin of such basic forces.
- 4.1.2 All turbine-engined aeroplanes, for which the individual certificate of airworthiness was first issued on or after 1 January 1987 but before 1 January 1989, with a maximum certificated take-off mass of over 27000kg that are of types of which the prototype was certificated by the appropriate national authority after 30 September, 1969 shall be equipped with an FDR which shall record at least first 16 parameters listed in table 1 of Appendix I.
- 4.1.3 All turbine-engined aeroplanes, for which the individual certificate of airworthiness was first issued on or after 1 January 1987 but before 1 January 1989, with a maximum certificated take-off mass of over 5700kg, except those in 4.1.2 should be equipped with an FDR which should record at least the first 9 parameters listed in table 1 of Appendix I.
- 4.1.4 All aeroplanes of a maximum certificated take-off mass of over 27000kg for which the individual certificate of airworthiness is first issued on or after 1 January 1989 shall be equipped with an FDR which shall record at least the first 32 parameters listed in table 1 of Appendix I.
- 4.1.5 All aeroplanes of a maximum certificated take-off mass of over 5700kg, up to and including 27000kg, for which the individual certificate of Airworthiness is first issued on or after 1 January 1989, shall be equipped with an FDR which shall record at least the first 16 parameters listed in table 1 of Appendix I.
- 4.1.6 All aeroplanes of a maximum certificated take-off mass of over 5700kg for which the individual certificate of airworthiness is first issued after 1 January 2005 shall be equipped with an FDR which shall record at least the first 78 parameters listed in table 1 of Appendix I.
- 4.1.7 All multi-engined turbine powered aeroplanes of a maximum certificated

take-off mass of 5700kg or less for which the individual Certificate of airworthiness is first issued on or after 1 January 1990, should be equipped with an FDR which should record at least the first 16 parameters listed in table 1 of Appendix I.

4.1.8 All turbine-engined aeroplanes, for which the individual certificate of airworthiness was first issued before 1 January 1989, with a maximum certificated take-off mass of over 5 700 kg, except those in 4.1.2, shall be equipped with an FDR which shall record at least the first 5 parameters listed in table 1 of Appendix I.

4.1.9 All turbine-engined aeroplanes of a maximum certificated take-off mass of 5700kg or less for which the application for type certification is submitted on or after 1 January 2016 shall be equipped with:

- a) an FDR which shall record at least the first 16 parameters listed in Table 1 of Appendix I; or
- b) a Class C AIR or AIRS which shall record at least the flight path and speed parameters displayed to the pilot(s), as defined in 2.1.2.3 of Appendix I; or
- c) an ADRS which shall record at least the first 7 parameters defined listed in Table-4 of Appendix I.

**Note(1)-** *“The application for type certification is submitted refers to the date of application of the original “Type Certificate” for the aero plane type, not the date of certification of particular aeroplane variants or derivative models.*

**Note(2)-** *AIR or AIRS classification is defined in 6.6.2.*

4.1.10 All turbine-engined aeroplanes of a maximum certificated take-off mass of 5700kg or less for which the individual certificate of airworthiness is first issued on or after 1 January 2016 should be equipped with:

- a) an FDR which should record at least the first 16 parameters listed in Table 1 of Appendix I; or
- b) a Class C AIR or AIRS which should record at least the flight path and speed parameters displayed to the pilot(s), as defined in 2.1.2.3 of Appendix I; or
- c) an ADRS which should record at least the first 7 parameters listed in table 4 of Appendix I.

4.1.11 All aeroplanes of a maximum certificated take-off mass of over 5 700 kg for which the application for type certification is submitted on or after 1<sup>st</sup> January, 2023 shall be equipped with an FDR capable of recording at least the 82 parameters listed in Table-1 of Appendix-I.

4.1.12 All aeroplanes of a maximum certificated take-off mass of over 5700 kg for which the individual certificate of airworthiness is first issued on or after 1<sup>st</sup> January, 2023 shall be equipped with an FDR capable of recording at least the 82 parameters listed in Table-1 of Appendix-I

#### 4.2 **Aeroplane- General Aviation**

4.2.1 All aeroplanes of a maximum certificated take-off mass of over 5700 kg for which the individual certificate of airworthiness is first issued on or after 1<sup>st</sup> January, 2005 shall be equipped with an FDR which shall record at least first 78 parameters listed in Table I of Appendix I.

4.2.2 All aeroplanes of a maximum certificated take-off mass of over 27000 kg for which the individual certificate of airworthiness is first issued on or after 1<sup>st</sup> January, 1989 shall be equipped with an FDR which shall record at least first 32 parameters listed in Table I of Appendix I.

4.2.3 All aeroplanes of a maximum certificated take-off mass of over 5700 kg, up to and including 27000 kg, for which the individual certificate of airworthiness is first issued on or after 1<sup>st</sup> January, 1989, should be equipped with an FDR which shall record at least first 16 parameters listed in Table I of Appendix I.

4.2.4 All multi-engined turbine powered aeroplanes of a maximum certificated takeoff mass of 5700kg or less for which the individual certificate of airworthiness is first issued on or after 1<sup>st</sup> January, 1990, should be equipped with an FDR which should record at least the first 16 parameters listed in Table 1 of Appendix I.

4.2.5 All turbine-engined aeroplanes with a seating configuration of more than five passenger seats and a maximum certificated take-off mass of 5700 kg or less for which the individual certificate of airworthiness is first issued on or after 1 January 2016 should be equipped with:

- a) an FDR which should record at least the first 16 parameters in Table-1 of Appendix-I; or
- b) a Class C AIR or AIRS which should record at least the flight path and speed parameters displayed to the pilot(s), as defined in 2.1.2.2 of Appendix-I; or
- c) an ADRS which should record at least first 7 parameters listed in Table-

4 of Appendix-I.

*Note-AIR or AIRS classification is defined in Para 6.6.*

- 4.2.6 All aeroplanes of a maximum certificated take-off mass of over 5 700 kg for which the application for type certification is submitted to a DGCA on or after 1<sup>st</sup> January, 2023 shall be equipped with an FDR capable of recording at least the 82 parameters listed in Table-1 of Appendix-I.
- 4.2.7 All aeroplanes of a maximum certificated take-off mass of over 5700 kg for which the individual certificate of airworthiness is first issued on or after 1<sup>st</sup> January, 2023 should be equipped with an FDR capable of recording at least the 82 parameters listed in Table-1 of Appendix-I.

**Note** *“The application for type certification that is submitted to a contracting state” refers to the date of application of the original “Type Certificate” for the aeroplane type, not the date of certification of particular aeroplane variants or derivative models.*

4.3 **Helicopter- Commercial Air Transport & General Aviation**

- 4.3.1 All helicopters of a maximum certificated take-off mass of over 7000kg, or having a passenger seating configuration of more than nineteen, for which the individual certificate of airworthiness is first issued on or after 1<sup>st</sup> January, 1989 shall be equipped with an FDR which shall record at least the first 30 parameters in Table-2 of Appendix I.
- 4.3.2 All helicopters of a maximum certificated take-off mass of over 3175 kg, up to and including 7000kg, for which the individual certificate of airworthiness is first issued on or after 1 January 1989, shall be equipped with a FDR which should record at least the first 15 parameters in Table-2 of Appendix I;
- 4.3.3 All helicopters of a maximum certificated take-off mass of over 3175kg for which the individual certificate of airworthiness is first issued on or after 1 January 2016 shall be equipped with an FDR which shall record at least the first 48 parameters listed in Table-2 of Appendix I.
- 4.3.4 All turbine-engined helicopters of a maximum certificated take-off mass of over 2250 kg, up to and including 3175 kg for which the application for type certification was submitted to a contracting state on or after 1 January 2018 shall be equipped with:
- a) an FDR which shall record at least the first 48 parameters in Table-2 of Appendix I; or
  - b) a Class C AIR or AIRS which shall record at least the flight path and

speed parameters displayed to the pilot(s), as defined in Table 5 of Appendix I; or

- c) an ADRS which shall record the first 7 parameters listed in Table 5 of Appendix I.

**Note** - *The “application for type certification was submitted to a contracting state refers to the date of application of the original “Type Certificate” for the helicopter type, not the date of certification of particular helicopter variants or derivative models.*

- 4.3.5 All helicopters of a maximum certificated take-off mass of 3175 kg or less for which the individual certificate of airworthiness is first issued on or after 1<sup>st</sup> January 2018 should be equipped with:

- a) an FDR which should record at least the first 48 parameters listed in Table-2 of Appendix I; or
- b) a Class C AIR or AIRS which should record at least the flight path and speed parameters displayed to the pilot(s), as defined in Table 5 of Appendix I; or
- c) an ADRS which should record the first 7 parameters listed in Table 5 of Appendix I;

**Note.** *AIR or AIRS classification is defined in Para 6.6.*

- 4.3.6 All helicopters of a maximum certificated take-off mass of over 3175kg for which the application for type certificate is submitted to a contracting state on or after 1<sup>st</sup> January 2023 shall be equipped with an FDR capable of recording at least the first 53 parameters listed in table 2 of Appendix I.

- 4.3.7 All helicopters of a maximum certificated take-off mass of over 3175kg for which the individual certificate of airworthiness is first issued on or after 1<sup>st</sup> January, 2023 shall be equipped with an FDR capable of recording at least the first 53 parameters listed in table 2 of Appendix I.

#### 4.4 **Recording Technology**

FDRs, ADRS, AIRs or AIRS shall not use engraving metal foil, frequency modulation (FM), photographic film or magnetic tape.

#### 4.5 **Duration**

- 4.5.1 All FDRs (installed on aeroplanes) shall retain the information recorded during at least the last 25 hours of their operation with the exception of those installed on aeroplane referred in 4.1.7 for which the FDR shall retain the

information recorded during at least the last 30 minutes of its operations and in addition sufficient information from the preceding take off for calibration purpose.

- 4.5.2 All FDRs (installed on helicopters) shall retain the information recorded during at least the last 10 hours of their operation.

## 5. COMBINATION RECORDERS

- 5.1 Combination recorders (FDR/CVR) may be used to meet the flight recorder equipage requirements in this CAR.
- 5.2 All aeroplanes of a maximum certificated take-off mass of over 5700kg for which the application for type certification is submitted to a contracting state on or after 1 January 2016, and which are required to be equipped with both a CVR and an FDR, should be equipped with two combination recorders (FDR/CVR).
- 5.3 All aeroplanes of a maximum certificated take-off mass of over 15000kg for which the application for type certification is submitted to a contracting state on or after 1 January 2016, and which are required to be equipped with both a CVR and an FDR, shall be equipped with two combination recorders (FDR/CVR). One recorder shall be located as close to the cockpit as practicable and the other recorder located as far aft as practicable.
- 5.4 All aeroplanes of a maximum certificated take-off mass over 5700kg, required to be equipped with an FDR and a CVR, may alternatively be equipped with two combination recorders (FDR/CVR).

**Note-** *The requirement of 5.1 5.2, 5.3 and 5.4 may be satisfied by equipping the aeroplanes with two combination recorders (one forward and one aft) or separate devices.*

- 5.5 All multi- engine turbine –powered aeroplanes of a maximum certificated take-off mass of 5700kg or less, required to be equipped with an FDR and a CVR, may alternatively be equipped with one combination recorder(FDR/CVR).

## 5.6 Automatic Deployable Flight Recorder (ADFR)

The following requirements shall apply to an ADFR:

- deployment shall take place when the aeroplane structure has been significantly deformed;
- deployment shall take place when an aeroplane sinks in water;
- ADFR shall not be capable of manual deployment;



- the ADFR shall be able to float on water;
- the ADFR deployment shall not compromise the safe continuation of the flight;
- the ADFR deployment shall not significantly reduce the chance of survival of the recorder and of successful transmission by its ELT;
- the ADFR deployment shall not release more than one piece;
- an alert shall be made to the flight crew when the ADFR is no longer captive to the aircraft.
- the flight crew shall have no means to disable ADFR deployment when the aircraft is airborne;
- the ADFR shall contain an integrated ELT, which shall activate automatically during the deployment sequence. Such ELT may be of a type that is activated in flight and provides information from which a position can be determined; and
- the integrated ELT of an ADFR shall satisfy the same requirements as an ELT required to be installed on an aeroplane. The integrated ELT shall at least have the same performance as the fixed ELT to maximize detection of the transmitted signal.

*Note 1 - Refer to the Manual on Location of Aircraft in Distress and Flight Recorder Data Recovery (Doc 10054) for more information on ADFR.*

*Note 2 - If an integrated ELT of a type that is activated in flight is used within an ADFR it could be a means to comply with requirements of Location of Aircraft in Distress (Doc 10054).*

## **6. DATA LINK RECORDERS**

### **6.1 Applicability**

6.1.1 All aeroplanes and helicopters for which the individual certificate of airworthiness are first issued on or after 1 January 2016, which use any of the data link communications applications referred to in para 6.4.2 and are required to carry a CVR, shall record the data link communications messages on a crash protected flight recorder.

6.1.2 All aeroplanes and helicopter for which the individual certificate of airworthiness was first issued before 1 January 2016, that are required to carry a CVR and are modified on or after 1 January 2016 to use any of the data link communications applications referred to in para 6.4.2 and shall record the data link communication messages on a crash protected flight recorder unless the installed data link communications equipment is compliant with a type

certificate issued or aircraft modification first approved prior to 1 January 2016.

**Note 1-** *A Class B AIR (Airborne Image Recorder) could be a means for recording data link communications applications messages to and from the aeroplanes/ helicopters where it is not practical or is prohibitively expensive to record those data link communications applications messages on FDR or CVR.*

**Note 2.—** *The “aircraft modifications” refer to modifications to install the data link communications equipment on the aircraft (e.g. structural, wiring).*

- 6.1.3 All aeroplanes and helicopters for which the individual certificate of airworthiness was first issued before 1 January 2016, that are required to carry a CVR and are modified on or after 1 January 2016 to use any of the data link communications applications referred to in para 6.4.2 should record the data link communications messages on a crash-protected flight recorder.

## **6.2 Duration**

The minimum recording duration shall be equal to the duration of the CVR.

## **6.3 Correlation**

Data link recording shall be able to be correlated to the recorded cockpit audio.

## **6.4 Applications to be recorded**

- 6.4.1 Where the aircraft/helicopter flight path is authorized or controlled through the use of data link messages, all data link messages, both uplinks (to the aircraft/helicopter) and downlinks (from the aircraft/helicopter), shall be recorded on the aircraft/helicopter. As far as practicable, the time the messages were displayed to the flight crew and the time of the responses shall be recorded.

- 6.4.2 Messages applying to the applications listed at Table-3 of Appendix-I shall be recorded. Applications without the asterisk (\*) are mandatory applications which shall be recorded regardless of the system complexity. Applications with an (\*) shall be recorded only as far as is practicable given the architecture of the system.

**Note-** *Sufficient information to derive the content of the data link communications message and the time the messages were displayed to the flight crew is needed to determine an accurate sequence of events on board the aircraft/helicopter.*

## **6.5 Flight Crew-Machine Interface Recordings**

### **6.5.1 Applicability**

6.5.1.1 All aeroplanes of a maximum take-off mass of over 27 000 kg for which the application for type certification is submitted on or after 1 January 2023 shall be equipped with a crash-protected flight recorder which shall record the information displayed to the flight crew from electronic displays, as well as the operation of switches and selectors by the flight crew as defined in Para 6.6.

6.5.1.2 All aeroplanes of a maximum take-off mass of over 5700 kg, up to and including 27 000 kg, for which the application for type certification is submitted on or after 1 January 2023 should be equipped with a crash-protected flight recorder which should record the information displayed to the flight crew from electronic displays, as well as the operation of switches and selectors by the flight crew, as defined in Para 6.6.

### **6.5.2 Duration**

The minimum flight crew-machine interface recording duration shall be at least for the last two hours.

### **6.5.3 Correlation**

Flight crew-machine interface recordings shall be able to be correlated to the recorded cockpit audio.

## **6.6 Start and stop logic**

6.6.1 The Airborne Image Recorder (AIR) and Airborne Image Recording System (AIRS) shall start to record prior to the aeroplane/ helicopter moving under its own power and record continuously until the termination of the flight when the aeroplane/ helicopter is no longer capable of moving under its own power. In addition, depending on the availability of electrical power, the AIR or AIRS shall start to record as early as possible during the cockpit checks prior to engine start at the beginning of the flight until the cockpit checks immediately following engine shutdown at the end of the flight.

### **6.6.2 Classes**

6.6.2.1 A Class “A” AIR or AIRS captures the general cockpit area in order to provide data supplemental to conventional flight recorders.

6.6.2.2 A Class “B” AIR or AIRS captures data link message displays.

6.6.2.3 A Class “C” AIR or AIRS captures instruments and control panels.

*Note 1-There are no provisions for Class “A” AIRs or AIRS in this CAR.*

*Note2- A Class C AIR or AIRS may be considered as a means for recording flight data where it is not practical or is prohibitively expensive to record on an FDR or an ADRS or where an FDR is not required.*

*Note3- To respect crew privacy, the cockpit area view may be designed as far as practical to exclude the head and shoulders of crew members whilst seated in their normal operating position.*

## **6.7 Applications to be recorded**

6.7.1 The operation of switches and selectors and the information displayed to the flight crew from electronic displays shall be captured by sensors or other electronic means.

6.7.2 The recording of operation of switches and selectors by the flight crew shall include the following:

- any switch or selector that will affect the operation and the navigation of the aircraft; and
- selection of normal and alternate systems.

6.7.3 The recording of the information displayed to the flight crew from electronic displays shall include the following:

- primary flight and navigation displays;
- aircraft system monitoring displays;
- engine indication displays;
- traffic, terrain, and weather displays;
- crew alerting systems displays;
- stand-by instruments; and
- installed EFB to the extent it is practical.

6.7.4 If image sensors are used, the recording of such images shall not capture the head and shoulders of the flight crew members whilst seated in their normal operating position

## **7. Flight Recorder Data Recovery**

- 7.1 All aeroplanes of a maximum certificated take-off mass of over 27000 kg and authorized to carry more than nineteen passengers for which the application for type certification is submitted to a Contracting State on or after 1 January 2021, shall be equipped with a means approved by the State of the Operator, to recover flight recorder data and make it available in a timely manner.
- 7.2 In approving the means to make flight recorder data available in a timely manner, the State of the Operator shall take into account the following:
- a) the capabilities of the operator;
  - b) overall capability of the aeroplane and its systems as certified by the State of Design;
  - c) the reliability of the means to recover the appropriate CVR channels and appropriate FDR data; and
  - d) specific mitigation measures.

Note— Guidance on approving the means to make flight recorder data available in a timely manner is contained in the Manual on Location of Aircraft in Distress and Flight Recorder Data Recovery (Doc 10054).

## **8. FLIGHT RECORDERS — GENERAL**

### **8.1 Construction and Installation**

- 8.1.1 Flight recorders shall be constructed, located and installed so as to provide maximum practical protection for the recordings in order that the recorded information may be preserved, recovered and transcribed. Flight recorders shall meet the prescribed crashworthiness and fire protection specifications.

### **8.2 Operation**

- 8.2.1 Flight recorders shall not be switched off during flight time.
- 8.2.2 To preserve flight recorder records, flight recorders shall be deactivated upon completion of flight time following an accident or incident. The flight recorders shall not be reactivated before their disposition as determined in accordance with the instructions issued by DGCA.

*Note 1 - The need for removal of the flight recorder records from the aircraft will be determined by DGCA with due regard to the seriousness of an occurrence and the circumstances, including the impact on the operation.*

*Note2 - The operator's responsibilities regarding the retention of flight recorder records are contained in 8.5.*

### **8.3 Continued Serviceability**

Operational checks and evaluations of recordings from the flight recorder systems shall be conducted to ensure the continued serviceability of the recorders.

*Note- Procedure for inspection of flight recorder systems are given in Appendix-I.*

### **8.4 Flight Recorder Electronic Documentation**

It is recommended that the documentation requirement concerning FDR and ADRS parameters provided by operators to accident investigation authorities should be in electronic format and take account of industry specifications.

*Note- Industry specification for documentation concerning flight recorder parameters may be found in the ARINC 647A, Flight Recorder Electronic Documentation, or equivalent document.*

### **8.5 Flight Recorder Records**

The pilot in-command, and/or owner/operator, shall ensure, to the extent possible, in the event the aeroplane/ helicopter becomes involved in an accident or incident, the preservation of all related flight recorder records and, if necessary, the associated flight recorders, and their retention in safe custody pending their disposition in accordance with instruction issued by DGCA.



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**APPENDIX- I**

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**GUIDANCE ON FLIGHT RECORDERS**

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**1. GENERAL REQUIREMENTS**

1.1 Non-deployable flight recorder containers shall be painted a distinctive orange colour.

1.2 Non-deployable crash-protected flight recorder containers shall:

- a) carry reflective material to facilitate their location; and
- b) have securely attached an automatically activated underwater locating device operating at a frequency of 37.5 kHz. At the earliest practicable date but not later than 1 January 2018, this device shall operate for a minimum of 90 days.

1.3 Automatic deployable flight recorder containers shall:

- a) be painted a distinctive orange colour, however the surface visible from outside the aircraft may be of another colour;
- b) carry reflective material to facilitate their location; and
- c) have an integrated automatically activated ELT.

1.4 The flight recorder systems shall be installed so that:

- a) the probability of damage to the recordings is minimized;
- b) there is an aural or visual means for pre-flight checking that the flight recorder systems are operating properly; and
- c) if the flight recorder systems have an erasure device, the installation shall be designed to prevent operation of the device during flight time or crash impact; and
- d) for aeroplanes for which the individual certificate of airworthiness is first issued on or after 1 January 2023, a flight crew-operated erase function shall be provided on the flight deck which, when activated, modifies the recording of a CVR and AIR so that it cannot be retrieved using normal replay or copying techniques. The installation shall be designed to prevent activation during

flight. In addition, the probability of an inadvertent activation of an erase function during an accident shall also be minimized.

**Note—** *The erase function is intended to prevent access to CVR and AIR recordings by normal replay or copying means, but would not prevent accident investigation authorities access to such recordings by specialized replay or copying techniques.*

1.5 The crash protected flight recorders shall be installed so that they receive electrical power from a bus that provides the maximum reliability for operation of the flight recorders without jeopardizing service to essential or emergency loads.

1.6 The lightweight flight recorders shall be connected to a power source having the characteristics which ensure proper and reliable recording in the operational environment.

1.7 The flight recorder systems, when tested by methods approved by the appropriate certificating authority, shall be demonstrated to be suitable for the environmental extremes over which they are designed to operate.

1.8 Means shall be provided for an accurate time correlation between the flight recorder systems recordings.

1.9 The Flight recorder system manufacturer shall provide the appropriate certificating authority with the following information in respect of the flight recorder systems:

- a) manufacturer's operating instructions, equipment limitations and installation procedures;
- b) parameter origin or source and equations which relate counts to units of measurement; and
- c) manufacturer's test reports; and
- d) detailed information to ensure the continued serviceability of the flight recorder system

1.10 The holder of the airworthiness approval for the installation design of the flight recorder system shall make available the relevant continuing airworthiness information to the operator of the aeroplane to be incorporated in the continuing airworthiness maintenance programme. This continuing airworthiness information shall cover in detail all the tasks required to ensure the continued serviceability of the flight recorder system.

**Note 1.—** *The flight recorder system is composed of the flight recorder as well as any dedicated sensors, hardware and software that provide*



*information required per this Appendix.*

*Note 2.— Conditions related to the continued serviceability of a flight recorder system are defined in Section 4 of this Appendix. The Manual on Flight Recorder System Maintenance (FRSM) (Doc 10104) provides guidance on maintenance tasks associated with flight recorder systems.*

## **2. FLIGHT DATA RECORDER (FDR) AND AIRCRAFT DATA RECORDING SYSTEMS (ADRS)**

### **2.1 Aeroplane:-**

#### **2.1.1 Start and stop logic**

The FDR or ADRS shall start to record prior to the aeroplane moving under its own power and record continuously until the termination of the flight when the aeroplane is no longer capable of moving under its own power.

#### **2.1.2 Parameters to be recorded**

2.1.2.1 The parameters that satisfy the requirements for FDRs are listed in the table 1 below. The number of parameters to be recorded shall depend on aeroplane complexity. The parameters without an asterisk (\*) are mandatory parameters which shall be recorded regardless of aeroplane complexity. In addition, the parameters designated by an asterisk (\*) shall be recorded if an information data source for the parameter is used by aeroplane systems or the flight crew to operate the aeroplane. However, other parameters may be substituted with due regard to the aeroplane type and the characteristics of the recording equipment.

2.1.2.2 If further FDR recording capacity is available, recording of the following additional information shall be considered:

- a) Operational information from electronic display systems, such as electronic flight instrument systems (EFIS), electronic centralized aircraft monitor (ECAM) and engine indication and crew alerting system (EICAS). Use the following order of priority:
  - 1) parameters selected by the flight crew relating to the desired flight path, e.g. barometric pressure setting, selected altitude, selected airspeed, decision height, and auto flight system engagement and mode indications if not recorded from another source;
  - 2) display system selection/status, e.g. SECTOR, PLAN, ROSE, NAV, WXR, COMPOSITE, COPY, ETC.;

3) warnings and alerts; and

4) the identity of displayed pages for emergency procedures and checklists; and

b) retardation information including brake application for use in the investigation of landing overruns and rejected take-offs.

2.1.2.3 The parameters that satisfy the requirements for flight path and speed as displayed to the pilot(s) are listed below. The parameters without an (\*) are mandatory parameters which shall be recorded. In addition, the parameters designated by an (\*) shall be recorded if an information source for the parameter is displayed to the pilot and is practicable to record:

- Pressure altitude
- Indicated airspeed or calibrated airspeed
- Heading (primary flight crew reference)
- Pitch attitude
- Roll attitude
- Engine thrust/power
- Landing-gear status\*
- Total or outside air temperature\*
- Time\*
- Navigation data\*: drift angle, wind speed, wind direction, latitude/longitude
- Radio altitude\*

2.1.2.4 The parameters that satisfy the requirements for ADRS are listed in Table 4.

2.1.2.5 The measurement range, recording interval and accuracy of parameters on installed equipment shall be verified by methods approved by the appropriate certificating authority.

2.1.2.6 Documentation concerning parameter allocation, conversion equations, periodic calibration and other serviceability/maintenance information shall be maintained by the operator. The documentation needs to be sufficient to ensure that accident investigation authorities have the necessary information to read out the data in engineering units.

## **2.2 Helicopters**

### **2.2.1 Start and stop logic**

The FDR or ADRS shall start to record prior to the helicopter moving under

its own power and record continuously until the termination of the flight when the helicopter is no longer capable of moving under its own power.

## **2.2.2 Parameters to be recorded**

2.2.2.1 The parameters that satisfy the requirements for FDRs, are listed in Table 2. The number of parameters to be recorded shall depend on helicopter complexity. The parameters without an asterisk (\*) are mandatory parameters which shall be recorded regardless of helicopter complexity. In addition, the parameters designated by an asterisk (\*) shall be recorded if an information data source for the parameter is used by helicopter systems or the flight crew to operate the helicopter. However, other parameters may be substituted with due regard to the helicopter type and the characteristics of the recording equipment.

2.2.2.2 The following parameters shall satisfy the requirements for flight path and speed:

- Pressure altitude
- Indicated airspeed
- Outside air temperature
- Heading
- Normal acceleration
- Lateral acceleration
- Longitudinal acceleration (body axis)
- Time or relative time count
- Navigation data\*: drift angle, wind speed, wind direction, latitude/longitude
- Radio altitude\*.

2.2.2.3 If further FDR recording capacity is available, recording of the following additional information shall be considered:

- a) additional operational information from electronic displays, such as electronic flight instrument systems (EFIS), electronic centralized aircraft monitor (ECAM) and engine indication and crew alerting system (EICAS); and
- b) additional engine parameters (EPR, N1, fuel flow, etc.).

2.2.2.4 The parameters that satisfy the requirements for ADRS are listed in Table 5.

2.2.2.5 The measurement range, recording interval and accuracy of parameters on installed equipment is usually verified by methods

approved by the appropriate certificating authority.

2.2.2.6 Documentation concerning parameter allocation, conversion equations, periodic calibration and other serviceability/maintenance information shall be maintained by the operator/owner. The documentation shall be sufficient to ensure that accident investigation authorities have the necessary information to read out the data in engineering units

### **3. RECORDING INTERVAL**

The flight recorders shall start to record prior to the aeroplane/helicopter moving under its own power and record continuously until the termination of the flight when the aeroplane/helicopter is no longer capable of moving under its own power.

### **4. INSPECTION OF FLIGHT RECORDER SYSTEM**

4.1 Prior to the first flight of the day, the built-in test features for the flight recorders and flight data acquisition unit (FDAU), when installed, shall be monitored by manual and/or automatic checks.

4.2 FDR systems or ADRS and AIR systems or AIRS shall have recording inspection intervals of one year; subject to the approval from the appropriate regulatory authority, this period may be extended to two years provided these systems have demonstrated a high integrity of serviceability and self-monitoring. DLR systems or DLRS shall have recording inspection intervals of two years; subject to the approval from the appropriate regulatory authority, this period may be extended to four years provided these systems have demonstrated high integrity of serviceability and self-monitoring.

4.3 Recording inspections shall be carried out as follows:

- a) an analysis of the recorded data from the flight recorders shall ensure that the recorder operates correctly for the nominal duration of the recording;
- b) The FDR or ADRS recording from a complete flight shall be examined in engineering units to evaluate the validity of all recorded parameters. Particular attention shall be given to parameters from sensors dedicated to the FDR or ADRS. Parameters taken from the aircraft's electrical bus system need not be checked if their serviceability can be detected by other aircraft systems;
- c) the readout facility shall have the necessary software to accurately convert the recorded values to engineering units and to determine the status of discrete signals;

- d) an examination of the recorded signal on the CVR or the CARS shall be carried out by replay of the CVR or CARS recording. While installed in the aircraft, the CVR or CARS shall record test signals from each aircraft source and from relevant external sources to ensure that all required signals meet intelligibility standards;
- e) where practicable, during the examination, a sample of in-flight recordings of the CVR or CARS shall be examined for evidence that the intelligibility of the signal is acceptable; and
- f) an examination of the recorded images on the AIR or AIRS shall be carried out by replay of the AIR or AIRS recording. While installed in the aircraft, the AIR or AIRS shall record test images from each aircraft source and from relevant external sources to ensure that all required images meet recording quality standards.
- g) an examination of the recorded messages on the DLR or DLRS shall be carried out by replay of the DLR or DLRS recording.

4.4 A flight recorder system shall be considered unserviceable if there is a significant period of poor quality data, unintelligible signals, or if one or more of the mandatory parameters is not recorded correctly.

4.5 A report of the recording system inspection shall be made available on request to DGCA for monitoring purposes.

**4.6 Calibration of the FDR system:**

- a) for those parameters which have sensors dedicated only to the FDR and are not checked by other means, recalibration shall be carried out at an interval determined by the continuing airworthiness information for the FDR system. In the absence of such information, a recalibration shall be carried out at least every five years. The recalibration shall determine any discrepancies in the engineering conversion routines for the mandatory parameters and to ensure that parameters are being recorded within the calibration tolerances; and
- b) When the parameters of altitude and airspeed are provided by sensors that are dedicated to the FDR system, there shall be a recalibration performed at an interval determined by the continuing airworthiness information for the FDR system. In the absence of such information, a recalibration shall be carried out at least every two years.

**Table-1**

**PARAMETER CHARACTERISTICS FOR FLIGHT DATA RECORDERS AEROPLANES**

| Serial number | Parameter                                                                  | Applicability                                                               | Measurement range                                                                                 | Maximum sampling and recording interval (seconds) | Accuracy limits (sensor input compared to FDR read-out)       | Recording resolution                                                  |
|---------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------|
| 1             | Time (UTC when available, otherwise relative time count or GNSS time sync) |                                                                             | 24 hours                                                                                          | 4                                                 | $\pm 0.125\%$ /h                                              | 1s                                                                    |
| 2             | Pressure-altitude                                                          |                                                                             | -300 m (-1 000 ft) to maximum certificated altitude of aircraft<br>+1 500 m (+5 000 ft)           | 1                                                 | $\pm 30$ m to $\pm 200$ m ( $\pm 100$ ft to $\pm 700$ ft)     | 1.5 m (5 ft)                                                          |
| 3             | Indicated airspeed or calibrated airspeed                                  |                                                                             | 95 km/h (50 kt) to max V <sub>So</sub> (Note 1)<br>V <sub>So</sub> to 1.2 V <sub>D</sub> (Note 2) | 1                                                 | $\pm 5\%$<br>$\pm 3\%$                                        | 1 kt (0.5 kt recommended)                                             |
| 4             | Heading (primary flight crew reference)                                    |                                                                             | 360°                                                                                              | 1                                                 | $\pm 2^\circ$                                                 | 0.5°                                                                  |
| 5             | Normal acceleration (Note 8)                                               | Application for type certification is submitted on or before 1 January 2016 | -3 g to +6 g                                                                                      | 0.125                                             | $\pm 1\%$ of maximum range excluding datum error of $\pm 5\%$ | 0.004 g                                                               |
|               |                                                                            | Application for type certification is submitted on or after 1 January 2016  | -3 g to +6 g                                                                                      | 0.0625                                            | $\pm 1\%$ of maximum range excluding datum error of $\pm 5\%$ | 0.004 g                                                               |
| 6             | Pitch attitude                                                             |                                                                             | $\pm 75^\circ$ or usable range whichever is greater                                               | 0.25                                              | $\pm 2^\circ$                                                 | 0.5°                                                                  |
| 7             | Roll attitude                                                              |                                                                             | $\pm 180^\circ$                                                                                   | 0.25                                              | $\pm 2^\circ$                                                 | 0.5°                                                                  |
| 8             | Radio transmission keying                                                  |                                                                             | On-off (one discrete)                                                                             | 1                                                 |                                                               |                                                                       |
| 9             | Power on each engine (Note 4 -3)                                           |                                                                             | Full range                                                                                        | 1 (per engine)                                    | $\pm 2\%$                                                     | 0.2% of full range or the resolution required to operate the aircraft |
| 10*           | Trailing edge flap and cockpit control selection                           |                                                                             | Full range or each discrete position                                                              | 2                                                 | $\pm 5\%$ or as pilot's indicator                             | 0.5% of full range or the resolution required to operate the aircraft |
| 11*           | Leading edge flap and cockpit control selection                            |                                                                             | Full range or each discrete position                                                              | 2                                                 | $\pm 5\%$ or as pilot's indicator                             | 0.5% of full range or the resolution required to operate the aircraft |
| 12*           | Thrust reverser position                                                   |                                                                             | Stowed, in transit,                                                                               | 1 (per engine)                                    |                                                               |                                                                       |

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| Serial number | Parameter                                                                                                | Applicability                                                                                  | Measurement range                     | Maximum sampling and recording interval (seconds) | Accuracy limits (sensor input compared to FDR read-out)                                      | Recording resolution                                                                     |
|---------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|               |                                                                                                          |                                                                                                | and reverse                           |                                                   |                                                                                              |                                                                                          |
| 13*           | Ground spoiler/speed brake selection (selection and position)                                            |                                                                                                | Full range or each discrete position  | 1                                                 | ±2% unless higher accuracy uniquely required                                                 | 0.2% of full range                                                                       |
| 14            | Outside air temperature                                                                                  |                                                                                                | Sensor range                          | 2                                                 | ±2°C                                                                                         | 0.3°C                                                                                    |
| 15*           | Autopilot/auto throttle/AFCS mode and engagement status                                                  |                                                                                                | A suitable combination of discretises | 1                                                 |                                                                                              |                                                                                          |
| 16            | Longitudinal acceleration ( <i>Note 8</i> )                                                              | Application for type certification submitted before 1 January 2016                             | ±1 g                                  | 0.25                                              | ±0.015 g excluding a datum error of ±0.05 g                                                  | 0.004 g                                                                                  |
|               |                                                                                                          | Application for type certification submitted on or after 1 January 2016                        | ±1 g                                  | 0.0625                                            | ±0.015 g excluding a datum error of ±0.05 g                                                  | 0.004 g                                                                                  |
| 17            | Lateral acceleration ( <i>Note 8</i> )                                                                   | Application for type certification submitted to a Contracting State before 1 January 2016      | ±1 g                                  | 0.25                                              | ±0.015 g excluding a datum error of ±0.05 g                                                  | 0.004 g                                                                                  |
|               |                                                                                                          | Application for type certification submitted to a Contracting State on or after 1 January 2016 | ±1 g                                  | 0.0625                                            | ±0.015 g excluding a datum error of ±0.05 g                                                  | 0.004 g                                                                                  |
| 18            | Pilot input and/or control surface position-primary controls (pitch, roll, yaw) ( <i>Notes 4 and 8</i> ) | Application for type certification submitted before 1 January 2016                             | Full range                            | 0.25                                              | ±2° unless higher accuracy uniquely required                                                 | 0.2% of full range or as installed                                                       |
|               |                                                                                                          | Application for type certification submitted on or after 1 January 2016                        | Full range                            | 0.125                                             | ±2° unless higher accuracy uniquely required                                                 | 0.2% of full range or as installed                                                       |
| 19            | Pitch trim position                                                                                      |                                                                                                | Full range                            | 1                                                 | ±3% unless higher accuracy uniquely required                                                 | 0.3% of full range or as installed                                                       |
| 20*           | Radio altitude                                                                                           |                                                                                                | –6 m to 750 m (–20 ft to 2 500 ft)    | 1                                                 | ±0.6 m (±2 ft) or ±3% whichever is greater below 150 m (500 ft) and ±5% above 150 m (500 ft) | 0.3 m (1 ft) below 150 m (500 ft) 0.3 m (1 ft) + 0.5% of full range above 150 m (500 ft) |
| 21*           | Vertical beam deviation (ILS/ GNSS/GLS glide path, MLS elevation, IRNAV/IAN vertical deviation)          |                                                                                                | Signal range                          | 1                                                 | ±3%                                                                                          | 0.3% of full range                                                                       |
| 22*           | Horizontal beam deviation (ILS/ GNSS/GLS localizer, MLS azimuth, IRNAV/IAN lateral deviation)            |                                                                                                | Signal range                          | 1                                                 | ±3%                                                                                          | 0.3% of full range                                                                       |
| 23            | Marker beacon passage                                                                                    |                                                                                                | Discrete                              | 1                                                 |                                                                                              |                                                                                          |
| 24            | Master warning                                                                                           |                                                                                                | Discrete                              | 1                                                 |                                                                                              |                                                                                          |

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| Serial number | Parameter                                                                                                                                                                                          | Applicability                                                                                                   | Measurement range                                      | Maximum sampling and recording interval (seconds) | Accuracy limits (sensor input compared to FDR read-out) | Recording resolution                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------|----------------------------------------|
| 25            | Each NAV receiver frequency selection ( <i>Note 5</i> )                                                                                                                                            |                                                                                                                 | Full range                                             | 4                                                 | As installed                                            |                                        |
| 26*           | DME 1 and 2 distance (includes Distance to runway threshold (GLS) and Distance to missed approach point (IRNAV/IAN)) ( <i>Notes 5 and 6</i> )                                                      |                                                                                                                 | 0 – 370 km<br>(0 – 200 NM)                             | 4                                                 | As installed                                            | 1 852 m (1 NM)                         |
| 27            | Air/ground status                                                                                                                                                                                  |                                                                                                                 | Discrete                                               | 1                                                 |                                                         |                                        |
| 28*           | GPWS/TAWS/GCAS status (selection of terrain display mode including pop-up display status) and (terrain alerts, both cautions and warnings, and advisories) and (on/off switch position)            |                                                                                                                 | Discrete                                               | 1                                                 |                                                         |                                        |
| 29*           | Angle of attack                                                                                                                                                                                    |                                                                                                                 | Full range                                             | 0.5                                               | As installed                                            | 0.3 % of full range                    |
| 30*           | Hydraulics, each system (low pressure)                                                                                                                                                             |                                                                                                                 | Discrete                                               | 2                                                 |                                                         | 0.5% of full range                     |
| 31*           | Navigation data (latitude/longitude, ground speed and drift angle) ( <i>Note 7</i> )                                                                                                               |                                                                                                                 | As installed                                           | 1                                                 | As installed                                            |                                        |
| 32*           | Landing gear and gear selector position                                                                                                                                                            |                                                                                                                 | Discrete                                               | 4                                                 | As installed                                            |                                        |
| 33*           | Groundspeed                                                                                                                                                                                        |                                                                                                                 | As installed                                           | 1                                                 | Data should be obtained from the most accurate system   | 1 kt                                   |
| 34            | Brakes (left and right brake pressure, left and right brake pedal position)                                                                                                                        |                                                                                                                 | (Maximum metered brake range, discretes or full range) | 1                                                 | ±5%                                                     | 2% of full range                       |
| 35*           | Additional engine parameters (EPR, N <sub>1</sub> , indicated vibration level, N <sub>2</sub> , EGT, fuel flow, fuel cut-off lever position, N <sub>3</sub> , engine fuel metering valve position) | Engine fuel metering valve position: Application for type certification is submitted on or after 1 January 2023 | As installed                                           | Each engine each second                           | As installed                                            | 2% of full range                       |
| 36*           | TCAS/ACAS (traffic alert and collision avoidance system)                                                                                                                                           |                                                                                                                 | Discrete                                               | 1                                                 | As installed                                            |                                        |
| 37*           | Wind shear warning                                                                                                                                                                                 |                                                                                                                 | Discrete                                               | 1                                                 | As installed                                            |                                        |
| 38*           | Selected barometric setting (pilot, co-pilot)                                                                                                                                                      |                                                                                                                 | As installed                                           | 64                                                | As installed                                            | 0.1 mb (0.01 in-Hg)                    |
| 39*           | Selected altitude (all pilot selectable modes of operation)                                                                                                                                        |                                                                                                                 | As installed                                           | 1                                                 | As installed                                            | Sufficient to determine crew selection |
| 40*           | Selected speed (all pilot selectable modes of operation)                                                                                                                                           |                                                                                                                 | As installed                                           | 1                                                 | As installed                                            | Sufficient to determine crew selection |
| 41*           | Selected Mach (all pilot selectable modes of operation)                                                                                                                                            |                                                                                                                 | As installed                                           | 1                                                 | As installed                                            | Sufficient to determine crew selection |



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| Serial number | Parameter                                                                                                                  | Applicability | Measurement range | Maximum sampling and recording interval (seconds) | Accuracy limits (sensor input compared to FDR read-out) | Recording resolution                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|---------------------------------------------------|---------------------------------------------------------|----------------------------------------|
| 42*           | Selected vertical speed (all pilot selectable modes of operation)                                                          |               | As installed      | 1                                                 | As installed                                            | Sufficient to determine crew selection |
| 43*           | Selected heading (all pilot selectable modes of operation)                                                                 |               | As installed      | 1                                                 | As installed                                            | Sufficient to determine crew selection |
| 44*           | Selected flight path (all pilot selectable modes of operation) (course/DSTRK, path angle, final approach path (IRNAV/IAN)) |               |                   | 1                                                 | As installed                                            |                                        |
| 45*           | Selected decision height                                                                                                   |               | As installed      | 64                                                | As installed                                            | Sufficient to determine crew selection |
| 46*           | EFIS display format (pilot, co-pilot)                                                                                      |               | Discrete(s)       | 4                                                 | As installed                                            |                                        |
| 47*           | Multi-function/engine/alerts display format                                                                                |               | Discrete(s)       | 4                                                 | As installed                                            |                                        |
| 48*           | AC electrical bus status                                                                                                   |               | Discrete(s)       | 4                                                 | As installed                                            |                                        |
| 49*           | DC electrical bus status                                                                                                   |               | Discrete(s)       | 4                                                 | As installed                                            |                                        |
| 50*           | Engine bleed valve position                                                                                                |               | Discrete(s)       | 4                                                 | As installed                                            |                                        |
| 51*           | APU bleed valve position                                                                                                   |               | Discrete(s)       | 4                                                 | As installed                                            |                                        |
| 52*           | Computer failure                                                                                                           |               | Discrete(s)       | 4                                                 | As installed                                            |                                        |
| 53*           | Engine thrust command                                                                                                      |               | As installed      | 2                                                 | As installed                                            |                                        |
| 54*           | Engine thrust target                                                                                                       |               | As installed      | 4                                                 | As installed                                            | 2% of full range                       |
| 55*           | Computed centre of gravity                                                                                                 |               | As installed      | 64                                                | As installed                                            | 1% of full range                       |
| 56*           | Fuel quantity in CG trim tank                                                                                              |               | As installed      | 64                                                | As installed                                            | 1% of full range                       |
| 57*           | Head up display in use                                                                                                     |               | As installed      | 4                                                 | As installed                                            |                                        |
| 58*           | Para visual display on/off                                                                                                 |               | As installed      | 1                                                 | As installed                                            |                                        |
| 59*           | Operational stall protection, stick shaker and pusher activation                                                           |               | As installed      | 1                                                 | As installed                                            |                                        |
| 60*           | Primary navigation system reference (GNSS, INS, VOR/DME, MLS, Loran C, localizer glideslope)                               |               | As installed      | 4                                                 | As installed                                            |                                        |
| 61*           | Ice detection                                                                                                              |               | As installed      | 4                                                 | As installed                                            |                                        |
| 62*           | Engine warning each engine vibration                                                                                       |               | As installed      | 1                                                 | As installed                                            |                                        |
| 63*           | Engine warning each engine over temperature                                                                                |               | As installed      | 1                                                 | As installed                                            |                                        |
| 64*           | Engine warning each engine oil pressure low                                                                                |               | As installed      | 1                                                 | As installed                                            |                                        |
| 65*           | Engine warning each engine over speed                                                                                      |               | As installed      | 1                                                 | As installed                                            |                                        |
| 66*           | Yaw trim surface position                                                                                                  |               | Full range        | 2                                                 | ±3% unless higher accuracy uniquely required            | 0.3% of full range                     |
| 67*           | Roll trim surface position                                                                                                 |               | Full range        | 2                                                 | ±3% unless higher accuracy uniquely required            | 0.3% of full range                     |
| 68*           | Yaw or sideslip angle                                                                                                      |               | Full range        | 1                                                 | ±5%                                                     | 0.5°                                   |
| 69*           | De-icing and/or anti-icing systems selection                                                                               |               | Discrete(s)       | 4                                                 |                                                         |                                        |
| 70*           | Hydraulic pressure (each system)                                                                                           |               | Full range        | 2                                                 | ±5%                                                     | 100 psi                                |
| 71*           | Loss of cabin pressure                                                                                                     |               | Discrete          | 1                                                 |                                                         |                                        |
| 72*           | Cockpit trim control input position, Pitch                                                                                 | Full range    | 1                 | ±5%                                               | 0.2% of full range or as installed                      |                                        |

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| Serial number | Parameter                                                                             | Applicability                                                           | Measurement range                                                    | Maximum sampling and recording interval (seconds) | Accuracy limits (sensor input compared to FDR read-out) | Recording resolution               |
|---------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------|------------------------------------|
| 73*           | Cockpit trim control input position, Roll                                             |                                                                         | Full range                                                           | 1                                                 | ±5%                                                     | 0.2% of full range or as installed |
| 74*           | Cockpit trim control input position, Yaw                                              |                                                                         | Full range                                                           | 1                                                 | ±5%                                                     | 0.2% of full range or as installed |
| 75*           | All cockpit flight control input forces (control wheel, control column, rudder pedal) |                                                                         | Full range (±311 N (±70 lbf), ± 378 N (±85 lbf), ± 734 N (±165 lbf)) | 1                                                 | ±5%                                                     | 0.2% of full range or as installed |
| 76*           | Event marker                                                                          |                                                                         | Discrete                                                             | 1                                                 |                                                         |                                    |
| 77*           | Date                                                                                  |                                                                         | 365 days                                                             | 64                                                |                                                         |                                    |
| 78*           | ANP or EPE or EPU                                                                     |                                                                         | As installed                                                         | 4                                                 | As installed                                            |                                    |
| 79*           | Cabin pressure altitude                                                               | Application for type certification submitted on or after 1 January 2023 | As installed (0 ft to 40 000 ft recommended)                         | 1                                                 | As installed                                            | 100 ft                             |
| 80*           | Aeroplane computed weight                                                             | Application for type certification submitted on or after 1 January 2023 | As installed                                                         | 64                                                | As installed                                            | 1% of full range                   |
| 81*           | Flight director command                                                               | Application for type certification submitted on or after 1 January 2023 | Full range                                                           | 1                                                 | ± 2°                                                    | 0.5°                               |
| 82*           | Vertical speed                                                                        | Application for type certification submitted on or after 1 January 2023 | As installed                                                         | 0.25                                              | As installed (32 ft/min recommended)                    | 16 ft/min                          |

Notes.—

1.  $V_{so}$ stalling speed or minimum steady flight speed in the landing configuration is in Section “Abbreviations and Symbols”.
2.  $V_D$  design diving speed.
3. Record sufficient inputs to determine power.
4. For aeroplanes with control systems in which movement of a control surface will back drive the pilot’s control, “or” applies. For aeroplanes with control systems in which movement of a control surface will not back drive the pilot’s control, “and” applies. In aeroplanes with split surfaces, a suitable combination of inputs is acceptable in lieu of recording each surface separately. In aeroplanes with independent pilot input on primary controls, each pilot input on primary controls needs to be recorded separately.
5. If signal available in digital form.
6. Recording of latitude and longitude from INS or other navigation system is a preferred alternative.
7. If signals readily available.
8. It is not intended that aeroplanes issued with an individual certificate of airworthiness before 1 January 2016 be modified to meet the measurement range, maximum sampling and recording interval, accuracy limits or recording resolution description detailed in this Appendix.

**Table-2**  
**PARAMETER CHARACTERISTICS FOR FLIGHT DATA RECORDERS - HELICOPTER**

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| Serial number | Parameter                                                                                                                                   | Applicability | Measurement range                                                                  | Maximum sampling and recording interval (seconds) | Accuracy limits (sensor input compared to FDR read-out)   | Recording resolution            |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------|---------------------------------|
| 1             | Time (UTC when available, otherwise relative time count or GNSS time sync)                                                                  |               | 24 hours                                                                           | 4                                                 | $\pm 0.125\%$ /h                                          | 1 s                             |
| 2             | Pressure-altitude                                                                                                                           |               | -300 m (-1000 ft) to maximum certificated altitude of aircraft +1500 m (+5 000 ft) | 1                                                 | $\pm 30$ m to $\pm 200$ m ( $\pm 100$ ft to $\pm 700$ ft) | 1.5 m (5 ft)                    |
| 3             | Indicated airspeed                                                                                                                          |               | As installed pilot display measuring system                                        | 1                                                 | $\pm 3\%$                                                 | 1 kt (0.5 kt recommended)       |
| 4             | Heading                                                                                                                                     |               | 360°                                                                               | 1                                                 | $\pm 2^\circ$                                             | 0.5°                            |
| 5             | Normal acceleration (Note 8)                                                                                                                |               | -3 g to +6 g                                                                       | 0.125                                             | $\pm 0.09$ g excluding a datum error of $\pm 0.05$ g      | 0.004 g                         |
| 6             | Pitch attitude                                                                                                                              |               | $\pm 75^\circ$ or 100% of usable range whichever is greater                        | 0.5                                               | $\pm 2^\circ$                                             | 0.5°                            |
| 7             | Roll attitude                                                                                                                               |               | $\pm 180^\circ$                                                                    | 0.5                                               | $\pm 2^\circ$                                             | 0.5°                            |
| 8             | Radio transmission keying                                                                                                                   |               | On-off (one discrete)                                                              | 1                                                 |                                                           |                                 |
| 9             | Power on each engine (Note 4 -3)                                                                                                            |               | Full range                                                                         | 1 (per engine)                                    | $\pm 2\%$                                                 | 0.1% of full range              |
| 10            | Main Rotor                                                                                                                                  |               |                                                                                    |                                                   |                                                           |                                 |
|               | Main Rotor Speed                                                                                                                            |               | 50-130%                                                                            | 0.51                                              | $\pm 2\%$                                                 | 0.3% of full range              |
|               | Rotor Brake                                                                                                                                 |               | Discrete                                                                           |                                                   |                                                           |                                 |
| 11            | Pilot input and/or control surface – primary controls (collective pitch, longitudinal cyclic pitch, lateral cyclic pitch, tail rotor pedal) |               | Full range                                                                         | 0.5 (0.25 recommended)                            | $\pm 2\%$ unless higher accuracy uniquely required        | 0.5% of operating range         |
| 12            | Hydraulics, each system (low pressure and selection)                                                                                        |               | Discrete                                                                           | 1                                                 |                                                           |                                 |
| 13            | Outside air temperature                                                                                                                     |               | Sensor range                                                                       | 2                                                 | $\pm 2^\circ\text{C}$                                     | 0.3°C                           |
| 14*           | Autopilot/auto throttle/AFCS mode and engagement status                                                                                     |               | A suitable combination of discretises                                              | 1                                                 |                                                           |                                 |
| 15*           | Stability Augmentation System Engagement                                                                                                    |               | Discrete                                                                           | 1                                                 |                                                           |                                 |
| 16*           | Main Gearbox Oil Pressure                                                                                                                   |               | As installed                                                                       | 1                                                 | As installed                                              | 6.895 kN/m <sup>2</sup> (1 psi) |
| 17*           | Main Gearbox Oil Temperature                                                                                                                |               | As installed                                                                       | 2                                                 | As installed                                              | 1°C                             |
| 18            | Yaw Rate                                                                                                                                    |               | $\pm 400^\circ$ /second                                                            | 0.25                                              | $\pm 1.5\%$ maximum range                                 | $\pm 2^\circ$ /s                |

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| Serial number | Parameter                                                                                   | Applicability | Measurement range                          | Maximum sampling and recording interval (seconds) | Accuracy limits (sensor input compared to FDR read-out)                                                             | Recording resolution                                                                     |
|---------------|---------------------------------------------------------------------------------------------|---------------|--------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|               |                                                                                             |               |                                            |                                                   | excluding datum error of $\pm 5\%$                                                                                  |                                                                                          |
| 19*           | Sling Load Force                                                                            |               | 0 to 200% of Certified Load                | 0.5                                               | $\pm 3\%$ of maximum range                                                                                          | 0.5% for maximum certified load                                                          |
| 20            | Longitudinal acceleration                                                                   |               | $\pm 1$ g                                  | 0.25                                              | $\pm 0.015$ g excluding a datum error of $\pm 0.05$ g                                                               | 0.004 g                                                                                  |
| 21            | Lateral acceleration<br>(Note 8)                                                            |               | $\pm 1$ g                                  | 0.25                                              | $\pm 0.015$ g excluding a datum error of $\pm 0.05$ g                                                               | 0.004 g                                                                                  |
| 22*           | Radio altitude                                                                              |               | -6 m to 750 m<br>(-20 ft to 2500 ft)       | 1                                                 | $\pm 0.6$ m ( $\pm 2$ ft) or $\pm 3\%$ whichever is greater below 150 m (500 ft) and $\pm 5\%$ above 150 m (500 ft) | 0.3 m (1 ft) below 150 m (500 ft) 0.3 m (1 ft) + 0.5% of full range above 150 m (500 ft) |
| 23*           | Vertical beam deviation                                                                     |               | Signal range                               | 1                                                 | $\pm 3\%$                                                                                                           | 0.3% of full range                                                                       |
| 24*           | Horizontal beam deviation                                                                   |               | Signal range                               | 1                                                 | $\pm 3\%$                                                                                                           | 0.3% of full range                                                                       |
| 25            | Marker beacon passage                                                                       |               | Discrete                                   | 1                                                 |                                                                                                                     |                                                                                          |
| 26            | Warnings                                                                                    |               | Discrete (s)                               | 1                                                 |                                                                                                                     |                                                                                          |
| 27            | Each Navigation receiver frequency selection                                                |               | Sufficient to determine selected frequency | 4                                                 | As installed                                                                                                        |                                                                                          |
| 28*           | DME 1 and 2 distances                                                                       |               | 0 – 370 km<br>(0 – 200 NM)                 | 4                                                 | As installed                                                                                                        | 1 852 m (1 NM)                                                                           |
| 29*           | Navigation data (latitude/longitude, ground Speed, drift angle, wind speed, wind direction) |               | As installed                               | 2                                                 | As installed                                                                                                        | As installed                                                                             |
| 30*           | Landing gear and gear selector position                                                     |               | Discrete                                   | 4                                                 |                                                                                                                     |                                                                                          |
| 31*           | Engine Exhaust Gas Temperature ( $T_4$ )                                                    |               | As installed                               | 1                                                 | As installed                                                                                                        |                                                                                          |
| 32*           | Turbine Inlet Temperature (TIT/ITT)                                                         |               | As installed                               |                                                   | As installed                                                                                                        |                                                                                          |
| 33*           | Fuel Contents                                                                               |               | As installed                               | 4                                                 | As installed                                                                                                        |                                                                                          |
| 34*           | Altitude Rate                                                                               |               | As installed                               | 1                                                 | As installed                                                                                                        |                                                                                          |
| 35*           | Ice Detection                                                                               |               | As installed                               | 4                                                 | As installed                                                                                                        |                                                                                          |
| 36*           | Helicopter Health and Usage Monitor System                                                  |               | As installed                               |                                                   | As installed                                                                                                        |                                                                                          |
| 37            | Engine Control Modes                                                                        |               | Discrete                                   | 1                                                 |                                                                                                                     |                                                                                          |
| 38*           | Selected barometric setting (pilot, co-pilot)                                               |               | As installed                               | 64 (4 recommended)                                | As installed                                                                                                        | 0.1 mb (0.01 in-Hg)                                                                      |
| 39*           | Selected altitude (all pilot selectable modes of operation)                                 |               | As installed                               | 1                                                 | As installed                                                                                                        | Sufficient to determine crew selection                                                   |
| 40*           | Selected speed (all pilot selectable modes of operation)                                    |               | As installed                               | 1                                                 | As installed                                                                                                        | Sufficient to determine crew selection                                                   |
| 41*           | Selected Mach (all pilot selectable modes of operation)                                     |               | As installed                               | 1                                                 | As installed                                                                                                        | Sufficient to determine crew selection                                                   |

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| Serial number | Parameter                                                                                                                                                                                                          | Applicability                                                                                       | Measurement range | Maximum sampling and recording interval (seconds) | Accuracy limits (sensor input compared to FDR read-out) | Recording resolution                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------|---------------------------------------------------------|----------------------------------------|
| 42*           | Selected vertical speed (all pilot selectable modes of operation)                                                                                                                                                  |                                                                                                     | As installed      | 1                                                 | As installed                                            | Sufficient to determine crew selection |
| 43*           | Selected heading (all pilot selectable modes of operation)                                                                                                                                                         |                                                                                                     | As installed      | 1                                                 | As installed                                            | Sufficient to determine crew selection |
| 44*           | Selected flight path (all pilot selectable modes of operation)                                                                                                                                                     |                                                                                                     | As installed      | 1                                                 | As installed                                            | Sufficient to determine crew selection |
| 45*           | Selected decision height                                                                                                                                                                                           |                                                                                                     | As installed      | 4                                                 | As installed                                            | Sufficient to determine crew selection |
| 46*           | EFIS display format (pilot, co-pilot)                                                                                                                                                                              |                                                                                                     | Discrete(s)       | 4                                                 |                                                         |                                        |
| 47*           | Multi-function/engine/alerts display format                                                                                                                                                                        |                                                                                                     | Discrete(s)       | 4                                                 |                                                         |                                        |
| 48*           | Event Marker                                                                                                                                                                                                       |                                                                                                     | Discrete          | 1                                                 |                                                         |                                        |
| 49*           | GPWS/TAWS/GCAS status ( selection of terrain display modes including pop-up display status) and (Terrain Alerts, both cautions and warnings, and advisories) and (on/off switch position) and (Operational Status) | Application for Type Certification is submitted to a contracting state on or after 1 January, 2023) | Discrete(s)       | 1                                                 | As installed                                            |                                        |
| 50*           | TCAS/ACAS (Traffic Alert and Collision Avoidance System) and (Operational Status)                                                                                                                                  | Application for Type Certification is submitted on or after 1 January, 2023                         | Discrete(s)       | 1                                                 | As installed                                            |                                        |
| 51*           | Primary Flight controls - Pilot input forces                                                                                                                                                                       | Application for type certification is submitted on or after 1 January 2023                          | Full range        | 0.125 (0.0625 recommended)                        | ±3% unless higher accuracy is uniquely required         | 0.5% of operating range                |
| 52*           | Computed Centre of Gravity                                                                                                                                                                                         | Application for type certification is submitted on or after 1 January 2023                          | As installed      | 64                                                | As installed                                            | 1% of full range                       |
| 53*           | Helicopter Computed Weight                                                                                                                                                                                         | Application for type certification is submitted on or after 1 January 2023                          | As installed      | 64                                                | As installed                                            | 1% of full range                       |

**Table-3**

**DESCRIPTION OF APPLICATIONS FOR DATA LINK RECORDERS**

| Item No. | Application type                      | Application description                                                                                                                                                                                                                                                                                                                                              | Recoding content |
|----------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.       | Data link Initiation                  | This includes any applications used to logon to or initiate data link service. In FANS-1/A and ATN, these are ATS Facilities Notification (AFN) and Context Management (CM) respectively.                                                                                                                                                                            | C                |
| 2.       | Controller/Pilot Communication        | This includes any application used to exchange requests, clearances, instructions and reports between the flight crew and controllers on the ground. In FANS-1/A and ATN, this includes the CPDLC application. It also includes applications used for the exchange of oceanic (OCL) and departure clearances (DCL) as well as data link delivery of taxi clearances. | C                |
| 3.       | Addressed Surveillance                | This includes any surveillance application in which the ground sets up contracts for delivery of surveillance data. In FANS-1/A and ATN, this includes the Automatic Dependent Surveillance - contract (ADS-C) application. Where parametric data are reported within the message they shall be recorded unless data from the same source are recorded on the FDR.   | C                |
| 4.       | Flight Information                    | This includes any service used for delivery of flight information to specific aircraft. This includes, for example, data link aviation weather report service (D-METAR), data link-automatic terminal service (D-ATIS), digital Notice to Airmen (D-NOTAM) and other textual data link services                                                                      | C                |
| 5.       | Aircraft Broadcast Surveillance       | This includes Elementary and Enhanced Surveillance Systems, as well as automatic dependent surveillance-broadcast (ADS-B) output data. Where parametric data sent by the helicopter are reported within the message they shall be recorded unless data from the same source are recorded on the FDR.                                                                 | M*               |
| 6.       | Aeronautical Operational Control Data | This includes any application transmitting or receiving data used for aeronautical operational control purposes (per the ICAO definition of operational control).                                                                                                                                                                                                    | M*               |

**Key:**

**C:** Complete contents recorded.

**M:** Information that enables correlation to any associated records stored separately from the helicopter.

**.\*:** Applications that are to be recorded only as far as is practicable given the architecture of the system.

**Table-4**

**PARAMETER CHARACTERISTICS FOR AIRCRAFT DATA RECORDING SYSTEMS  
- AEROPLANES**

| No. | Parameter name                | Minimum recording range                                                              | Maximum recording interval in seconds | Minimum recording accuracy                                             | Minimum recording resolution | Remarks                                                                     |
|-----|-------------------------------|--------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------|
| 1   | Heading                       |                                                                                      |                                       |                                                                        |                              |                                                                             |
|     | a) Heading (Magnetic or True) | ±180°                                                                                | 1                                     | ±2°                                                                    | 0.5°                         | Heading is preferred, if not available, yaw rate shall be recorded          |
|     | b) Yaw rate                   | ±300°/s                                                                              | 0.25                                  | ±1% + drift of 360°/h                                                  | 2°/s                         |                                                                             |
| 2   | a) Pitch attitude             | ±90°                                                                                 | 0.25                                  | ±2°                                                                    | 0.5°                         | Pitch attitude is preferred, if not available, pitch rate shall be recorded |
|     | b) Pitch rate                 | ±300°/s                                                                              | 0.25                                  | ±1% + drift of 360°/h                                                  | 2°/s                         |                                                                             |
| 3   | a) Roll attitude              | ±180°                                                                                | 0.25                                  | ±2°                                                                    | 0.5°                         | Roll attitude is preferred, if not available, roll rate shall be recorded   |
|     | b) Roll rate                  | ±300°/s                                                                              | 0.25                                  | ±1% + drift of 360°/h                                                  | 2°/s                         |                                                                             |
| 4   | Positioning system            |                                                                                      |                                       |                                                                        |                              |                                                                             |
|     | a) Time                       | 24 hours                                                                             | 1                                     | ±0.5 s                                                                 | 0.1 s                        | UTC time preferred where available.                                         |
|     | b) Latitude/longitude         | Latitude:±90°<br>Longitude:±180°                                                     | 2<br>(1 if available)                 | As installed<br>(0.00015° recommended)                                 | 0.00005°                     |                                                                             |
|     | c) Altitude                   | -300 m (-1 000 ft) to maximum certificated altitude of aeroplane +1 500 m (5 000 ft) | 2<br>(1 if available)                 | As installed<br>(±15 m (±50 ft) recommended)                           | 1.5 m (5 ft)                 |                                                                             |
|     | d) Ground speed               | 0-1 000 kt                                                                           | 2<br>(1 if available)                 | As installed<br>(±5 kt recommended)                                    | 1 kt                         |                                                                             |
|     | e) Track                      | 0-360°                                                                               | 2<br>(1 if available)                 | As installed<br>(± 2° recommended)                                     | 0.5°                         |                                                                             |
|     | f) Estimated error            | Available range                                                                      | 2<br>(1 if available)                 | As installed                                                           | As installed                 | Shall be recorded if readily available                                      |
| 5   | Normal acceleration           | -3 g to + 6 g (*)                                                                    | 0.25 (0.125 if available)             | As installed (± 0.09 g excluding a datum error of ±0.45 g recommended) | 0.004 g                      |                                                                             |
| 6   | Longitudinal acceleration     | ±1 g (*)                                                                             | 0.25 (0.125 if available)             | As installed (±0.015 g excluding a                                     | 0.004 g                      |                                                                             |

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| No. | Parameter name                                                                       | Minimum recording range                                                   | Maximum recording interval in seconds | Minimum recording accuracy                                                                                      | Minimum recording resolution        | Remarks                                                                                                                                                                                                              |
|-----|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                                                                      |                                                                           |                                       | datum error of $\pm 0.05$ g recommended)                                                                        |                                     |                                                                                                                                                                                                                      |
| 7   | Lateral acceleration                                                                 | $\pm 1$ g (*)                                                             | 0.25 (0.125 if available)             | As installed ( $\pm 0.015$ g excluding a datum error of $\pm 0.05$ g recommended)                               | 0.004 g                             |                                                                                                                                                                                                                      |
| 8   | External static pressure (or pressure altitude)                                      | 34.4 mb (3.44 in-Hg) to 310.2 mb (31.02 in-Hg) or available sensor range  | 1                                     | As installed ( $\pm 1$ mb (0.1 in-Hg) or $\pm 30$ m ( $\pm 100$ ft) to $\pm 210$ m ( $\pm 700$ ft) recommended) | 0.1 mb (0.01 in-Hg) or 1.5 m (5 ft) |                                                                                                                                                                                                                      |
| 9   | Outside air temperature (or total air temperature)                                   | $-50^{\circ}$ to $+90^{\circ}\text{C}$ or available sensor range          | 2                                     | As installed ( $\pm 2^{\circ}\text{C}$ recommended)                                                             | $1^{\circ}\text{C}$                 |                                                                                                                                                                                                                      |
| 10  | Indicated air speed                                                                  | As the installed pilot display measuring system or available sensor range | 1                                     | As installed ( $\pm 3$ % recommended)                                                                           | 1 kt (0.5 kt recommended)           |                                                                                                                                                                                                                      |
| 11  | Engine RPM                                                                           | Full range including overspeed condition                                  | Each engine each second               | As installed                                                                                                    | 0.2% of full range                  |                                                                                                                                                                                                                      |
| 12  | Engine oil pressure                                                                  | Full range                                                                | Each engine each second               | As installed (5% of full range recommended)                                                                     | 2% of full range                    |                                                                                                                                                                                                                      |
| 13  | Engine oil temperature                                                               | Full range                                                                | Each engine each second               | As installed (5% of full range recommended)                                                                     | 2% of full range                    |                                                                                                                                                                                                                      |
| 14  | Fuel flow or pressure                                                                | Full range                                                                | Each engine each second               | As installed                                                                                                    | 2% of full range                    |                                                                                                                                                                                                                      |
| 15  | Manifold pressure                                                                    | Full range                                                                | Each engine each second               | As installed                                                                                                    | 0.2% of full range                  |                                                                                                                                                                                                                      |
| 16  | Engine thrust/power/torque parameters required to determine propulsive thrust/power* | Full range                                                                | Each engine each second               | As installed                                                                                                    | 0.1% of full range                  | * Sufficient parameters e.g. EPR/N1 or torque/Np as appropriate to the particular engine shall be recorded to determine power in both normal and reverse thrust. A margin for possible overspeed should be provided. |
| 17  | Engine gas generator speed (Ng)                                                      | 0-150%                                                                    | Each engine each second               | As installed                                                                                                    | 0.2% of full range                  |                                                                                                                                                                                                                      |
| 18  | Free power turbine speed (Nf)                                                        | 0-150%                                                                    | Each engine each second               | As installed                                                                                                    | 0.2% of full range                  |                                                                                                                                                                                                                      |



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| No. | Parameter name                          | Minimum recording range              | Maximum recording interval in seconds | Minimum recording accuracy                          | Minimum recording resolution | Remarks                                                                |
|-----|-----------------------------------------|--------------------------------------|---------------------------------------|-----------------------------------------------------|------------------------------|------------------------------------------------------------------------|
| 19  | Coolant temperature                     | Full range                           | 1                                     | As installed ( $\pm 5^{\circ}\text{C}$ recommended) | 1° C                         |                                                                        |
| 20  | Main voltage                            | Full range                           | Each engine each second               | As installed                                        | 1 Volt                       |                                                                        |
| 21  | Cylinder head temperature               | Full range                           | Each cylinder each second             | As installed                                        | 2% of full range             |                                                                        |
| 22  | Flaps position                          | Full range or each discrete position | 2                                     | As installed                                        | 0.5°                         |                                                                        |
| 23  | Primary flight control surface position | Full range                           | 0.25                                  | As installed                                        | 0.2 % of full range          |                                                                        |
| 24  | Fuel quantity                           | Full range                           | 4                                     | As installed                                        | 1% of full range             |                                                                        |
| 25  | Exhaust gas temperature                 | Full range                           | Each engine each second               | As installed                                        | 2% of full range             |                                                                        |
| 26  | Emergency voltage                       | Full range                           | Each engine each second               | As installed                                        | 1 Volt                       |                                                                        |
| 27  | Trim surface position                   | Full range or each discrete position | 1                                     | As installed                                        | 0.3% of full range           |                                                                        |
| 28  | Landing gear position                   | Each discrete position*              | Each gear every two seconds           | As installed                                        |                              | * Where available, record up-and- locked and down-and- locked position |
| 29  | Novel/unique aircraft features          | As required                          | As required                           | As required                                         | As required                  |                                                                        |

**Table - 5**

**PARAMETER CHARACTERISTICS FOR AIRCRAFT DATA RECORDING SYSTEMS  
- HELICOPTERS**

| No. | Parameter name                | Minimum recording range                                                             | Maximum recording interval in seconds | Minimum recording accuracy                        | Minimum recording resolution | Remarks                                                                      |
|-----|-------------------------------|-------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------|------------------------------|------------------------------------------------------------------------------|
| 1   | Heading                       |                                                                                     |                                       |                                                   |                              |                                                                              |
|     | a) Heading (Magnetic or True) | ±180°                                                                               | 1                                     | ±2°                                               | 0.5°                         | * Heading is preferred, if not available, yaw rate shall be recorded         |
|     | b) Yaw rate                   | ±300°/s                                                                             | 0.25                                  | ±1% + drift of 360°/h                             | 2°/s                         |                                                                              |
| 2   | Pitch                         |                                                                                     |                                       |                                                   |                              |                                                                              |
|     | a) Pitch attitude             | ±90°                                                                                | 0.25                                  | ±2°                                               | 0.5°                         | *Pitch attitude is preferred, if not available, pitch rate shall be recorded |
|     | b) Pitch rate                 | ±300°/s                                                                             | 0.25                                  | ±1% + drift of 360°/h                             | 2°/s                         |                                                                              |
| 3   | Roll                          |                                                                                     |                                       |                                                   |                              |                                                                              |
|     | a) Roll attitude              | ±180°                                                                               | 0.25                                  | ±2°                                               | 0.5°                         | * Roll attitude is preferred, if not available, roll rate shall be recorded  |
|     | b) Roll rate                  | ±300°/s                                                                             | 0.25                                  | ±1% + drift of 360°/h                             | 2°/s                         |                                                                              |
| 4   | Positioning system:           |                                                                                     |                                       |                                                   |                              |                                                                              |
|     | a) Time                       | 24 hours                                                                            | 1                                     | ±0.5 s                                            | 0.1 s                        | UTC time preferred where available.                                          |
|     | b) Latitude /longitude        | Latitude:±90°<br>Longitude:±180°                                                    | 2 (1 if available)                    | As installed (0.00015° recommended)               | 0.00005°                     |                                                                              |
|     | c) Altitude                   | −300 m (−1 000 ft) to maximum certificated altitude of aeroplane +1500 m (5 000 ft) | 2 (1 if available)                    | As installed (±15 m (±50 ft) recommended)         | 1.5 m (5 ft)                 |                                                                              |
|     | d) Ground speed               | 0–1 000 kt                                                                          | 2 (1 if available)                    | As installed (±5 kt recommended)                  | 1 kt                         |                                                                              |
|     | e) Track                      | 0–360°                                                                              | 2 (1 if available)                    | As installed (± 2° recommended)                   | 0.5°                         |                                                                              |
|     | f) Estimated error            | Available range                                                                     | 2 (1 if available)                    | As installed                                      | As installed                 | Shall be recorded if readily available                                       |
| 5   | Normal acceleration           | −3 g to + 6 g                                                                       | 0.25 (0.125 if available)             | As installed (± 0.09 g excluding a datum error of | 0.004 g                      |                                                                              |

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| No. | Parameter name                                                                       | Minimum recording range                                                    | Maximum recording interval in seconds | Minimum recording accuracy                                                           | Minimum recording resolution         | Remarks                                                                                                                                              |
|-----|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                                                                      |                                                                            |                                       | ±0.05 g recommended)                                                                 |                                      |                                                                                                                                                      |
| 6   | Longitudinal acceleration                                                            | ±1 g                                                                       | 0.25 (0.125 if available)             | As installed (±0.015 g excluding a datum error of ±0.05 g recommended)               | 0.004 g                              |                                                                                                                                                      |
| 7   | Lateral acceleration                                                                 | ±1 g                                                                       | 0.25 (0.125 if available)             | As installed (±0.015 g excluding a datum error of ±0.05 g recommended)               | 0.004 g                              |                                                                                                                                                      |
| 8   | External static pressure (or pressure altitude)                                      | 34.4 hPa (1.02 in -Hg) to 310.2 hPa (9.16 in-Hg) or available sensor range | 1                                     | As installed (±1 hPa (0.3 in-Hg) or ±30 m (±100 ft) to ±210 m (±700 ft) recommended) | 0.1 hPa (0.03 in-Hg) or 1.5 m (5 ft) |                                                                                                                                                      |
| 9   | Outside air temperature (or total air temperature)                                   | -50° to +90°C or available sensor range                                    | 2                                     | As installed (±2°C recommended)                                                      | 1°C                                  |                                                                                                                                                      |
| 10  | Indicated air speed                                                                  | As the installed pilot display measuring system or available sensor range  | 1                                     | As installed (±3 % recommended)                                                      | 1 kt (0.5 kt recommended)            |                                                                                                                                                      |
| 11  | Main rotor speed (Nr)                                                                | 50% to 130% or available sensor range                                      | 0.5                                   | As installed                                                                         | 0.3% of full range                   |                                                                                                                                                      |
| 12  | Engine RPM (*)                                                                       | Full range including overspeed condition                                   | Each engine each second               | As installed                                                                         | 0.2% of full range                   | * for piston-engined helicopters                                                                                                                     |
| 13  | Engine oil pressure                                                                  | Full range                                                                 | Each engine each second               | As installed (5% of full range recommended)                                          | 2% of full range                     |                                                                                                                                                      |
| 14  | Engine oil temperature                                                               | Full range                                                                 | Each engine each second               | As installed (5% of full range recommended)                                          | 2% of full range                     |                                                                                                                                                      |
| 15  | Fuel flow or pressure                                                                | Full range                                                                 | Each engine each second               | As installed                                                                         | 2% of full range                     |                                                                                                                                                      |
| 16  | Manifold pressure (*)                                                                | Full range                                                                 | Each engine each second               | As installed                                                                         | 0.2% of full range                   | * for piston-engined helicopters                                                                                                                     |
| 17  | Engine thrust/power/torque parameters required to determine propulsive thrust/power* | Full range                                                                 | Each engine each second               | As installed                                                                         | 0.1% of full range                   | * Sufficient parameters e.g. EPR/N1 or torque/Np as Appropriate to the particular engine shall be recorded to determine power. A margin for possible |

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| No. | Parameter name                      | Minimum recording range              | Maximum recording interval in seconds | Minimum recording accuracy                          | Minimum recording resolution | Remarks                                                              |
|-----|-------------------------------------|--------------------------------------|---------------------------------------|-----------------------------------------------------|------------------------------|----------------------------------------------------------------------|
|     |                                     |                                      |                                       |                                                     |                              | overspeed should be provided. Only for turbine-engined helicopters   |
| 18  | Engine gas generator speed (Ng) (*) | 0-150%                               | Each engine each second               | As installed                                        | 0.2% of full range           | * Only for turbine-engined helicopters                               |
| 19  | Free power turbine speed (Nf)(*)    | 0-150%                               | Each engine each second               | As installed                                        | 0.2% of full range           | * Only for turbine-engined helicopters                               |
| 20  | Collective Pitch                    | Full range                           | 0.5                                   | As installed                                        | 0.1% of full range           |                                                                      |
| 21  | Coolant temperature                 | Full range                           | 1                                     | As installed ( $\pm 5^{\circ}\text{C}$ recommended) | 1 $^{\circ}\text{C}$         | * Only for piston-engined helicopters                                |
| 22  | Main voltage                        | Full range                           | Each engine each second               | As installed                                        | 1 Volt                       |                                                                      |
| 23  | Cylinder head temperature (*)       | Full range                           | Each cylinder each second             | As installed                                        | 2% of full range             | * Only for piston-engined helicopters                                |
| 24  | Fuel quantity                       | Full range                           | 4                                     | As installed                                        | 1% of full range             |                                                                      |
| 25  | Exhaust gas temperature             | Full range                           | Each engine each second               | As installed                                        | 2% of full range             |                                                                      |
| 26  | Emergency voltage                   | Full range                           | Each engine each second               | As installed                                        | 1 Volt                       |                                                                      |
| 27  | Trim surface position               | Full range or each discrete position | 1                                     | As installed                                        | 0.3% of full range           |                                                                      |
| 28  | Landing gear position               | Each discrete position*              | Each gear every two seconds           | As installed                                        |                              | * Where available, record up-and-locked and down-and-locked position |
| 29  | Novel/unique aircraft features      | As required                          | As required                           | As required                                         | As required                  |                                                                      |

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